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Given: $x = 1$, $y = 2$, $w = 0.5$, $b = 0$, $v = 1$, $c = 0$, learning rate 0.1
Model: $z = wx + b$, $h = \tanh(z)$, $\hat{y} = vh + c$, $J = 0.5\,(\hat{y} - y)^2$

1. Forward pass

$z = 0.5(1) + 0 = \mathbf{0.5000}$

$h = \tanh(0.5) = \mathbf{0.462117157}$

$\hat{y} = 1(h) + 0 = \mathbf{0.462117157}$

$J = 0.5\,(\hat{y} - 2)^2 = 0.5\,(-1.537882843)^2 = \mathbf{1.182541819}$

2. Chain rule, backwards

$\frac{dJ}{d\hat{y}} = \hat{y} - y = \mathbf{-1.537882843}$

$\frac{dJ}{dc} = \frac{dJ}{d\hat{y}} \cdot 1 = \mathbf{-1.537882843}$

$\frac{dJ}{dv} = \frac{dJ}{d\hat{y}} \cdot h = \mathbf{-0.710682047}$

$\frac{dJ}{dh} = \frac{dJ}{d\hat{y}} \cdot v = \mathbf{-1.537882843}$

$\frac{dh}{dz} = 1 - h^2 = \mathbf{0.786447733}$

$\frac{dJ}{dz} = \frac{dJ}{dh} \cdot (1 - h^2) = \mathbf{-1.209464475}$

$\frac{dJ}{dw} = \frac{dJ}{dz} \cdot x = \mathbf{-1.209464475}$

$\frac{dJ}{db} = \frac{dJ}{dz} \cdot 1 = \mathbf{-1.209464475}$

All gradients are negative: the prediction is below the target.

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3. One simultaneous update (all four use the ORIGINAL w, b, v, c)

w_1 = 0.5 - 0.1(-1.209464475) = **0.620946448**

b_1 = 0 - 0.1(-1.209464475) = **0.120946448**

v_1 = 1 - 0.1(-0.710682047) = **1.071068205**

c_1 = 0 - 0.1(-1.537882843) = **0.153788284**

New loss J_1 = **0.685773765** (was 1.182541819: decreased)

4. Gradient direction, in words

The gradient points toward increasing loss. Gradient descent moves each parameter the opposite way, so w, b, v and c all increase, which raises ŷ toward the target 2 and lowers J. The loss falls from 1.1825 to 0.6858 after one step.

5. Check against the C++ program (central differences, ε = 1e-6)

	analytic	finite difference
dJ/dw	-1.20946448	-1.20946448
dJ/db	-1.20946448	-1.20946448
dJ/dv	-0.71068205	-0.71068205
dJ/dc	-1.53788284	-1.53788284

Largest absolute error from the program: 1.12e-10 (limit 1e-5). Loss after 101 updates: 1.08e-20.

Sign-off

Name: _____ ID: _____ Date: _____

Signature: _____